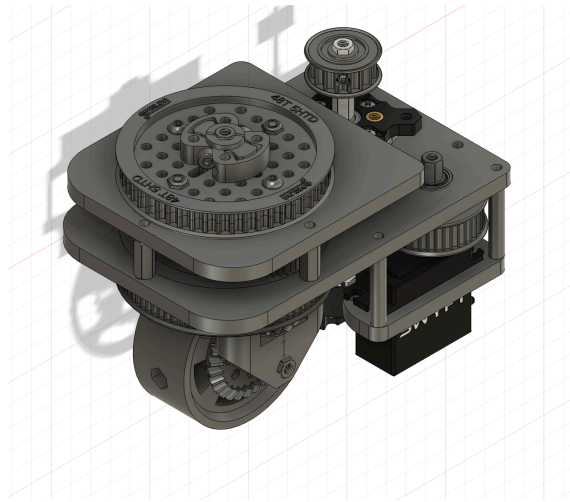


Coaxial Swerve Drivetrain Pod

Design & Hardware - Drivetrain Subsystem



Each pod of the drivetrain is a steerable drive module that lets a single wheel rotate in any direction while driving, achieving holonomic motion. The assembly packages the hardware between two plates to stay compact and rigid, with the steering handled by a belt driven pulley system, and drive routed through bevel gears and pulleys down to the wheel.

Frame & Structure

The pod is built on a sandwiched two-plate aluminum frame joined by standoffs. Stacking the plates on standoffs sets a gap for the steering and drive components, and gives the module stiffness without adding a heavy enclosure around it. All mounting uses the goBuilda hex hole/bore pattern, which keeps the plate layout, pulley mounting, and motor placement on a consistent grid.

Steering

The steering runs on a 81-tooth custom designed pulley that forms the rotation ring. A SWYFT motor drives this ring through a 5 mm HTD timing belt. The HTD belt

was chosen over a geared ring for lower friction, quieter steering and simpler tensioning; keeping the belt under tension is what limits backlash.

Drive

Drive power comes from a separate drive motor and is transmitted down through a pulley system and bevel gear set to the wheel. Routing the drive along the rotation axis keeps the drive path separate from steering, so rotating the module does not disturb wheel speed.

Key Specifications

Steering actuator	SWYFT servo
Steering ring	81 tooth custom designs pulley
Steering reduction	2.7:1
Drive motor	goBuilda bare motor with built in hex shaft
Drive transmission	Pulleys, and bevel gear set
Drive reduction	10.8:1 555 RPM
Wheel	72 mm / metal casted
Tread	Andimark Blue Nitrilite
Frame	Two-plate aluminum on standoffs
Primary hardware	goBuilda structural & motion components
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